

AVIN P S

ML Engineer / Data Scientist

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Summary

Final year Computer Science & Engineering student (Data Science) with hands-on experience in training and deploying production ML models, deep learning, Data Science and structured data extraction pipelines. Comfortable building models from scratch, running systematic experiments, preparing and validating labeled datasets, and writing clean automation scripts with a proven record of shipping work used in real-world settings.

Skills

Languages: Python (Advanced), SQL, JavaScript/TypeScript (Basic)

ML & Data Science: Scikit-Learn, Gradient Boosting, SHAP, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Predictive Modelling, Statistical Analysis, Model Evaluation, EDA

AI & NLP: HuggingFace Transformers, RAG, LangChain, ChromaDB, Gemini API, LLM Integration, Explainable AI, AI Decision Systems

Backend & APIs: Python, FastAPI, Pydantic, WebSocket, REST APIs, SQL, PostgreSQL, SQLAlchemy, Background Tasks

Frontend: React, TypeScript, JavaScript, Tailwind CSS, HTML / CSS, Recharts

DevOps & Tools: Git & GitHub, MLOps, Vercel, Vite, npm / pip, Render

Experience

ML Summer Intern | *Indian Institute of Information Technology, Kottayam*

May 2026 – Aug 2026

- **Built and evaluated ML models** (Logistic Regression, Decision Trees, Random Forest, XGBoost) across multiple datasets, applying SHAP, LIME, PDP and ICE for model interpretability
- **Designed an end-to-end sentiment analysis system** for 12,500+ Flipkart product reviews, using SHAP explanations to diagnose and fix a model bug, improving negative-class precision from 69 % to 87 %
- **Turned raw SHAP outputs into aspect-level, business-ready insights** (e.g. "customers complain about fit and stitching") by filtering for genuine product aspects rather than generic sentiment words
- **Deployed the full pipeline as an interactive web app** using Streamlit, enabling non-technical stakeholders to explore per-product sentiment and live predictions

Education

B.Tech Computer Science & Engineering (Data Science)

Graduation: 2027

APJ Abdul Kalam Technological University, IES College of Engineering, Thrissur, Kerala

CGPA: 8.1 / 10

Certifications

- **IBM Data Science Professional Certificate – Coursera (2025)**
- **Machine Learning Specialization Certificate – DeepLearning.AI & Stanford Online (2026)**
- **Mathematics for Machine Learning & Data Science Certificate – DeepLearning.AI (2026)**

Projects

AI Personal Health Coach | *Python, FastAPI, React, Scikit-Learn, SHAP, Gemini, PostgreSQL, Render* **Feb 2026 – Jul 2026**

- **Built and deployed 7 Gradient Boosting classifiers** end-to-end, from constructing a 75,000-sample synthetic dataset through feature engineering, hyperparameter tuning, and calibrated 0–100 risk scoring, served via a production FastAPI backend on Render with a React frontend.
- Engineered **per-sample SHAP (TreeExplainer)** explainability with semantic suppression logic, surfacing only each user's relevant, modifiable risk drivers and turning model outputs into transparent, plain-English explanations rather than black-box scores.
- Designed a **hybrid LLM recommendation engine** (Google Gemini + curated rule base) that generates grounded health guidance without fabricating medical claims, with deterministic fallbacks and PostgreSQL analytics persistence for full fault tolerance.

KTU Result Analyser | *React, TypeScript, PDF Parsing, Gemini API (LLM), LangChain, Vercel* **Dec 2025 – Feb 2026**

- **Built and deployed a production structured-data extraction pipeline** actively used by IES College of Engineering's management and also being distributed to multiple colleges: ingests raw, variably-formatted academic PDF documents and extracts structured fields student names, subject codes, grades, credits, and dates with zero manual intervention.
- Integrated a **NLP Large-Language Model (Gemini API)** for automated document understanding; engineered prompts for reliable structured output, handling edge cases in formatting and multi-page layouts, with downstream SGPA computation and downloadable cohort-level analytical reports.
- Demonstrates core skills directly relevant to document intelligence: **unstructured-to-structured extraction, field validation, and automation scripting** to eliminate repetitive manual data work.

SpaceX Falcon 9 Landing Prediction (IBM Capstone) | *Python, SQL, Scikit-Learn, Plotly Dash* **Jan 2026**

- **End-to-end ML pipeline** covering multi-source data collection (REST API + web scraping), ETL scripting, SQL-based EDA, feature engineering, and systematic multi-model comparison (Logistic Regression, SVM, KNN, Decision Tree) with documented hyperparameter tuning to **83% accuracy**.
- Ran **structured experiments**, tracked model performance across architectures, documented findings as technical reports, and built an interactive Plotly Dash dashboard for communicating results to non-technical stakeholders.

Leadership & Extracurricular

- **Chief Editor, DataXplorer** (Dept. Data Science Newsletter, 2024–Present) — curates and communicates ML research trends and AI developments; reflects strong written documentation habits and ability to explain technical work clearly.
- **Coordinator, College Technical & Arts Fests** – IES College of Engineering; managed cross-functional teams and end-to-end execution of college-wide events under high-pressure, time-bound conditions.